PREMKUMAR RAMESH

AUTOMATION ENGINEER|PYTHON & CAD AUTOMATION

Bengaluru, India| premramesh2001@gmail.com| 9003548094| www.linkedin.com/in/prem1147ramesh

SUMMARY

Automation Engineer with experience building API-driven, rule-based systems for CAD workflows. Specialized in configurable automation using SolidWorks API, Auto LISP, Visual LISP, Python, and Excel VBA to eliminate repetitive design tasks and scale engineering operations.

WORK EXPERIENCE

Associate Engineer - Automation

ACO Technology Centre **02/2025 - Present, Bengaluru**

- Designed and deployed **API-driven CAD automation pipelines** using SolidWorks API and AutoCAD Auto LISP to automatically generate configurable engineering drawings.
- Built parameter-based automation frameworks supporting geometry creation, validation, annotation, and standards enforcement across multiple product variants.
- Performed **design validation and DFMEA support**, ensuring automated outputs met engineering and quality standards.

Graduate Engineer Trainee - Automation

ACO Technology Centre **02/2024 - 02/2025, Bengaluru**

- Implemented **API-based automation** workflows to standardize drawing outputs and reduce manual drafting dependency across multiple product variants.
- Converted repetitive engineering logic into **reusable automation modules**, enabling configuration changes without manual CAD intervention.

Intern - Design

ACO Technology Centre **08/2023 - 02/2024, Bengaluru**

- Created CAD models and drawings in SolidWorks for drainage channels and grating systems across various load classes.
- Supported design, drafting and documentation activities for drainage channel and grating products.

SKILLS

CAD Automation & Programming: SolidWorks API, Auto LISP, Visual LISP, Python(automation scripting, data handling, API integration), Excel VBA
Design & CAD: SolidWorks, AutoCAD, Autodesk Fusion 360, Autodesk Inventor

ACADEMIC PROJECTS

Design and Fabrication of Soil Moisture content detecting rover with irrigation system

- Developed soil moisture monitoring rover with integrated irrigation and sensor-based data acquisition and testing systems.

Design and Fabrication of IoT-Based Sugarcane Harvesting Bot

- Developed IoT embedded harvesting prototype using SolidWorks for mechanical structure and system integration.
- Developed a control pipeline using **ESP32, ultrasonic sensors, and L293** motor driver for servo/DC motor actuation.

EDUCATION

Bachelor of Engineering

Mechatronics Engineering - Sri Krishna College of Engineering and Technology **09/2019 - 04/2023, Coimbatore**

KEY ACHIEVEMENTS

- Delivered rule-based CAD automation solutions using **AutoCAD Auto LISP and SolidWorks API**, reducing manual drafting effort by approximately **90%** and significantly improving drawing standardization.
- Automated engineering productivity reporting using Excel VBA, enabling centralized tracking and eliminating manual data entry previously performed by a **15**-member team.

LANGUAGES

English (Proficient), **Tamil** (Native)